

BSA Simple Guideline

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Revision History

Date	Version	Description	Author
2019/08/19	0.1	Description of how to run bsa_server , app_manager, app_ag, app_hs , app_avk and app_pbc	Leo Chien

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Initialization process

➤ BSA structure

One or more client applications can connect to the server to use Bluetooth services/profiles (FTP, AG, A2DP...etc.) The communication between clients and the servers is based on sockets and on named first in and first out(FIFO) pipes. Shown as Figure 1- BSA Structure

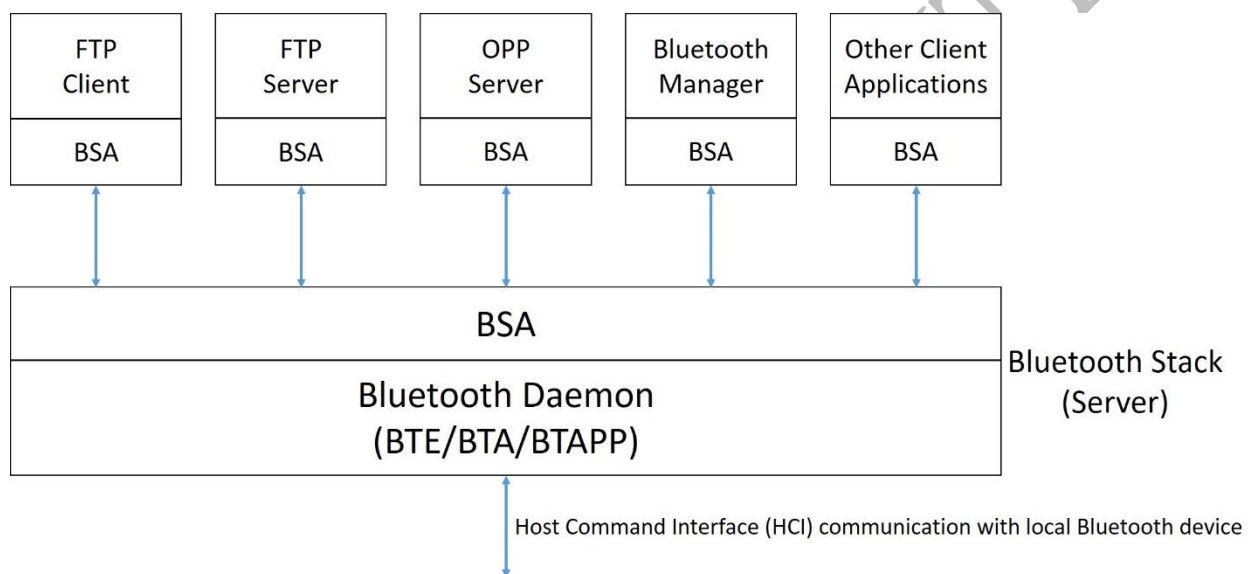


Figure 1- BSA Structure

➤ BSA compilation environment

For ARM platform, you will need to set cross compiler as below:

```
export ARMGCC=arm-openwrt-linux-gcc
```

```
export PATH=[yourtoolchain]/bin:$PATH
```

Please change your toolchain name and path accordingly.

➤ Compilation BSA sample application

To compile BSA sample program, please follow below steps:

1. `cd 3rdparty/embedded/bsa_examples/linux/app_manager/build`
2. `make CPU=arm clean`
3. `make CPU=arm`

You can compile others sample application in the same way.

`app_manager` – an application that can do inquiry and pairing procedure

`app_ag` - HFP AG role

`app_avk` - A2DP sink role

`app_hs` - HFP headset role

`app_pbc` - Phone Book Profile Client

`app_ble` – GATT

etc...

➤ **Execute the BSA application**

First of all, you should run `bsa_server/app_manager` and run others sample code depending on what profiles feature do you need.

1. RESET BT_REG_ON pin
2. Run `bsa_server`: `./bsa_server -d /dev/ttyS0 -p /path/to/xxx.hcd -r 14`

Please change `/dev/ttyS0`, BT firmware path and UART baudrate that platform can support accordingly.

`./bsa_server -help` view the options available to `bsa_server`. Shown as Figure 2 - `bsa_server` help menu

```

root@Bling:~/original/server/x86_64# ./bsa_server -help
USAGE:      bsa_server [OPTION]...

  -help          get option help information
  -b file        specify the BT Snoop file to save serial packets
  -d device      set the hci device interface (e.g. /dev/ttyHS0)
  -u path        prefix all the socket names with a path
  -p patchfile   force the patchfile (RAM) to use
  -r baudrate    set baud rate for opening port
                  Baud rate codes:
                  0  = 300 baud
                  1  = 600
                  2  = 1200
                  3  = 2400
                  4  = 9600
                  5  = 19200
                  6  = 57600
                  7  = 115200
                  8  = 230400
                  9  = 460800
                  10 = 921600
                  11 = 1M
                  12 = 1.5M
                  13 = 2M
                  14 = 3M
                  15 = 4M

  -all=#         set trace level to # for all layers
  -diag=#        set DIAG trace level to #
  -hci=#         set HCI trace level to #
  -l2c=#         set L2CAP trace level to #
  -app=#         set APPL trace level to #
  -btm=#         set BTM trace level to #
  -sdp=#         set SDP trace level to #
  -rfc=#         set RFCOMM trace level to #
  -gap=#         set GAP trace level to #
  -obx=#         set OBEX trace level to #
  -goep=#        set GOEP trace level to #
  -a2d=#         set A2DP trace level to #
  -avdt=#        set AVDT trace level to #
  -avct=#        set AVCT trace level to #
  -avrc=#        set AVRC trace level to #

```

Figure 2 - bsa_server help menu

app_manager

app_manager is mainly used to search and pair surrounding BT devices. Shown as Figure 3 - app_manager menu

To run application, you will need to put it in the sample folder as bsa_server as it will try to connect to the socket generated by bsa_server by default.(It could be changed by adding -u parameter to bsa_server to specify the socket location)

```
Bluetooth Application Manager Main menu:
 1 => Abort Discovery
 2 => Discovery
 3 => Discovery test
 4 => Bonding
 5 => Cancel Bonding
 6 => Remove device from security database
 7 => Services Discovery (all services)
 8 => Device Id Discovery
 9 => Set local Device Id
10 => Get local Device Id
11 => Stop Bluetooth
12 => Restart Bluetooth
13 => Accept Simple Pairing
14 => Refuse Simple Pairing
15 => Enter BLE Passkey
16 => Act As HID Keyboard (SP passkey entry)
17 => Read Device configuration
18 => Read Local Out Of Band data
19 => Enter remote Out Of Band data
20 => Set device visibility
21 => Set device BLE visibility
22 => Set AFH Configuration
23 => Set Tx Power Class2 (specific FW needed)
24 => Set Tx Power Class1.5 (specific FW needed)
25 => Change Dual Stack Mode (currently:DUAL_STACK_MODE_BSA)
26 => Set Link Policy
27 => Enter Passkey
28 => Get Remote Device Name
29 => RSSI Measurement
30 => Set class of device
96 => Kill BSA server
97 => Connect to BSA server
98 => Disconnect from BSA server
99 => Quit
```

Figure 3 - app_manager menu

Below are the steps to pair a device.

1. app_manager menu : input 2 (Discovery)

```
Select action => New Discovered device:0
Bdaddr:d4:0b:1a:0b:35:20
Name:Bling
ClassOfDevice:5a:02:0c => Phone
Services:0x00000000 ()
Rssi:-48
BSA_trace 57@ 01/01 23h:57m:12s:422ms: COMPLETE_LOCAL_NAME_TYPE:
BSA_trace 58@ 01/01 23h:57m:12s:422ms: 0000: 42 6c 69 6e 67 Bling
BSA_trace 59@ 01/01 23h:57m:12s:422ms: COMPLETE_16BITS_UUID_TYPE:
BSA_trace 60@ 01/01 23h:57m:12s:422ms: 0000: 05 11 06 11 0a 11 0c 11 0e 11 12 11 15 11 16 11 .....
BSA_trace 61@ 01/01 23h:57m:12s:422ms: 0010: 1f 11 2d 11 2f 11 00 12 32 11 .../...2.
BSA_trace 62@ 01/01 23h:57m:12s:422ms: COMPLETE_32BITS_UUID_TYPE:
BSA_trace 63@ 01/01 23h:57m:12s:422ms: 0000:
BSA_trace 64@ 01/01 23h:57m:12s:422ms: COMPLETE_128BITS_UUID_TYPE:
BSA_trace 65@ 01/01 23h:57m:12s:422ms: 0000:
Discovery complete
```

Discovery

Figure 4 - Discovery

2. app_manager menu : input 4 (Bonding)

```
Bluetooth Bond menu:
Dev:0
Bdaddr:d4:0b:1a:0b:35:20
Name:Bling
ClassOfDevice:5a:02:0c => Phone
Rssi:-40
Select device => 0
```

Bonding

Figure 5 - Bonding

3. input 13 (Accept Simple Pairing)

```
Select action => BSA_trace 51@ 01/01 08h:13m:42s:211ms: bsa_sec_event_hdlr event:0
DEBUG: app_mgr_security_callback: event:0
DEBUG: app_mgr_security_callback: BSA_SEC_LINK_UP_EVT bd_addr: d4:0b:1a:0b:35:20
DEBUG: app_mgr_security_callback: ClassOfDevice:5a:02:0c => Phone
BSA_trace 52@ 01/01 08h:13m:42s:451ms: bsa_sec_event_hdlr event:6
DEBUG: app_mgr_security_callback: event:6
DEBUG: app_mgr_security_callback: BSA_SEC_SP_CFM_REQ_EVT
DEBUG: app_mgr_security_callback: Remote device:Bling
DEBUG: app_mgr_security_callback: bd_addr: d4:0b:1a:0b:35:20
DEBUG: app_mgr_security_callback: ClassOfDevice:5a:02:0c => Phone
DEBUG: app_mgr_security_callback: Just Work:FALSE
DEBUG: app_mgr_security_callback: Numeric Value:295685
DEBUG: app_mgr_security_callback: You must accept or refuse using menu
```

Accept

Figure 6 - Accept

4. Successful pairing, bt_devices.xml record the paired devices

app_ag

app_ag is HFP AG role sample code. It can use to connect to headset and create SCO link. The menu is shown as Figure 7 - app_ag menu

```
Bluetooth Headset/HandsFree Main menu:
 1 => Discover Headset
 2 => Enable
 3 => Disable
 4 => Register
 5 => Deregister
 6 => Connect
 7 => Disconnect
 8 => Open audio
 9 => Close audio
10 => Record audio file
11 => Stop record audio file
12 => Play audio file
13 => Stop Playing audio file
14 => Display SCO route config
15 => Indicate incoming call
16 => PickUp call
17 => HangUp call
99 => Quit
Select action =>
```

Figure 7 - app_ag menu

Below is the sample to connect to a headset

1. app_ag menu : input 1 Discover headset
2. app_ag menu : input 6 (Connect)

```
Select action => 6
DEBUG: app_ag_open: Entering
Bluetooth AG menu:
    0 Device from XML database (already paired)
    1 Device found in last discovery
Select source => 0
Dev:0
    Bdaddr:00:09:dd:42:b9:91
    Name:SOUNDOT v1.0
    ClassOfDevice:00:00:00 => Misc device
    Trusted Services:0
    Link Key present:TRUE
    Pid:0 Vid:0
```

Figure 8 - app_ag connect

3. Input 0 to connect to device already paired or 1 to connect to device in last discovery
4. Select the BT device to be connected.

app_hs

app_ag is HFP client role sample code. It can acts as headset role to connect to HFP audio gateway. The sample code menu is Figure 9 - app_hs menu

```
Bluetooth Mono Headset Main menu:
 1 => discover
 2 => Connect
 3 => Close
 4 => pick up the call
 5 => hang up the call
 6 => Play audio file
 7 => Record audio file
 8 => Stop recording audio file
99 => Quit
Select action => █
```

Figure 9 - app_hs menu Follow the process below:

Below is the sample to connect to audio gateway

1. app_hs menu : input 2 (Connect)

```
Select action => 2
DEBUG: app_hs_open: Entering
Bluetooth AG menu:
 0 Device from XML database (already paired)
 1 Device found in last discovery
Select source => 0
Dev:0
  Bdaddr:b0:19:c6:8e:ba:df
  Name:Ming-Fu 的 iPhone
  ClassOfDevice:00:00:00 => Misc device
  Trusted Services:18
  Link Key present:TRUE
  Pid:0 Vid:0
```

Figure 10 - app_hs menu

2. Input 0 (Device from XML database).
3. Select the BT device to be connected, input 0.

```
BSA_HS_CONN_EVT:
- Remote bdaddr: d4:0b:1a:0b:35:20
- Service: Handsfree
```

Figure 11 - app_hs BSA_HS_CONN_EVT

4. select 4(pick up the call) to pick up when there is incoming call
5. select 5(hang up the call) to hang up

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app_avk

app_avk can implement A2DP sink role. App_avk menu is shown as Figure 12 - app_avk menu

```
Bluetooth AVK Main menu:
 1 => Start Discovery
 2 => AVK Register
 3 => AVK DeRegister
 4 => AVK Open
 5 => AVK Close
 6 => AVK Start steam play
 7 => AVK Stop stream play
 8 => AVK Pause stream play
 9 => AVK Play next track
10 => AVK Play previous track
11 => AVRC Command
12 => AVK Send Get Element Attributes Command
13 => AVK Get capabilities
14 => AVK Register Notification
15 => AVK Send Delay Report
99 => Quit
Select action => █
```

Figure 12 - app_avk menu

Follow the process below:

1. App_avk menu : input 2 (AVK Register)
2. App_avk menu : input 4 (AVK Open)

```
Select action => 4
Bluetooth AVK Open menu:
 0 Device from XML database (already paired)
 1 Device found in last discovery
Select source => 0
DEBUG: app_read_xml_remote_devices: read(./bt_devices.xml): OK
Dev:0
    Bdaddr:d4:0b:1a:0b:35:20
    Name:Bling
    ClassOfDevice:5a:02:0c => Phone
    Trusted Services:0
    Link Key present:TRUE
    Pid:0 Vid:0
Select device => 0
```

Figure 13 - app_avk open

3. Input 0 (Device from XML database)

4. Select the BT device to be connected, input 0

When it's connected, it will create a wav file to save the audio data coming from a2dp source side. We can modify the code to write the data to ALSA device for playback.

```
DEBUG: app_avk_cback: BSA_AVK_START_EVT status 0x0  
AVK start: format PCM  
DEBUG: app_wav_create_file: created WAV file test avk-1.wav  
Sampling rate:44100, number of channel:2 bit per sample:16  
BSA_trace 31@ 01/01 11h:44m:49s:878ms: bsa_avk_event_hdlr message_id:741  
BSA_trace 32@ 01/01 11h:44m:49s:878ms: bsa_avk_event_hdlr event:18  
DEBUG: app_avk_cback: BSA_AVK_CP_INFO_EVT
```

Figure 14 - app_avk create wav file

5. Do control app_avk stream play/pause/ next tract from app_avk menu